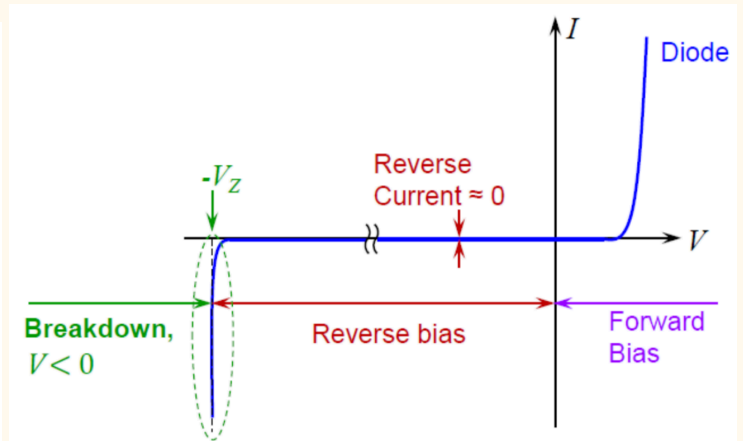
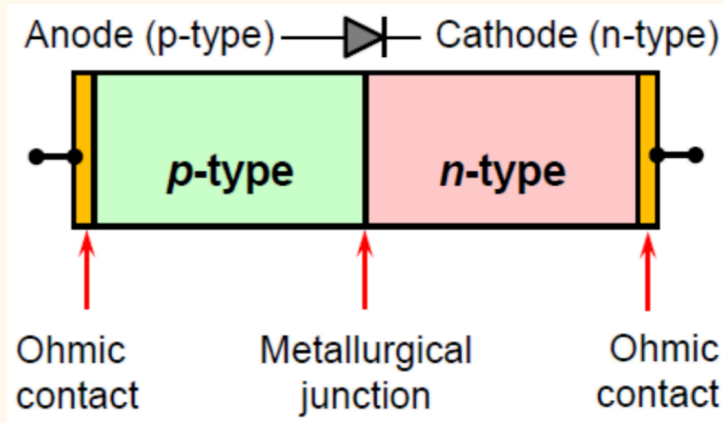


## Diode Summary

### Operating Regions and IV Characteristics



- CANNED APPLE
  - cathode is n
  - anode is p

| IV Characteristics                                                        |                                                                                                                                                                                                                                                                                            |
|---------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Substantially Forward Bias<br>$V_D \gg V_{cut-in}$<br>or<br>$I_D \gg I_S$ | $I_D = I_S \left( e^{\frac{V_D}{nV_T}} - 1 \right) \approx I_S e^{\frac{V_D}{nV_T}}$ $V_D = nV_T \ln\left(\frac{I_D}{I_S}\right) = nV_T \ln(I_D) - nV_T \ln(I_S)$ <ul style="list-style-type: none"> <li>• <math>V_D</math> and <math>\ln(I_D)</math> has a linear relationship</li> </ul> |
| Reverse Bias                                                              | $I_D = I_S \left( e^{\frac{V_D}{nV_T}} - 1 \right) \approx -I_S$                                                                                                                                                                                                                           |
| Reverse Breakdown                                                         | $V_D = V_Z$                                                                                                                                                                                                                                                                                |
| Diode Parameters                                                          |                                                                                                                                                                                                                                                                                            |
| Thermal Voltage                                                           | $V_T = \frac{kT_{Kelvin}}{q} \quad V_T \approx 25 \text{ mV} @ T = 300 \text{ K} (298 \text{ K}) = 25^\circ\text{C}$ $k = \text{Boltzmann's constant} = 1.380649 \times 10^{-23} \text{ J/K}$ $q = \text{elementary charge} = 1.60217663 \times 10^{-19} \text{ C}$                        |
| Exponential/Ideality Factor                                               | $n$ <p>1 = ideal, reality 1 ~ 2</p>                                                                                                                                                                                                                                                        |
| Reverse Saturation Current                                                | $I_S$                                                                                                                                                                                                                                                                                      |